



Digital IC Design & Verification
Physical Design
Project | Physical Design of RISC-V Single Cycle Processor

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Revision History

Revision Number	Revision Date	Revision By	Nature of Revision	Approved By
1.0	02/05/2025	Hira Sohail	Lab Manual	Sharjeel Khilji
2.0	09/10/2025	Fawad Ahmed	Additional Task	Sharjeel Khilji

Lab Task

Take RTL of your single-cycle processor and perform synthesis without DFT and then pass it through the full backend flow, generating a final GDSII file. The goal is to complete all steps with no violations and generate a comprehensive, well-documented report.

Note:

- Remove data memory and instruction memory from your designs and connect them from outside with your main core.
- Now, do complete back-end flow of your core without memories. Memories are made using SRAM / DRAM macros on our chip and we will deal with macros later.
- You can perform PD with Pins Placements only but it is preferred to use IO Pads.
- If anyone done PD with I/O Pads, **he/she will be awarded 1% extra weightage.**

1. Synthesize your netlist

- Perform gate-level netlist of your RTL design without DFT. Perform post- synthesis simulation and compare results with pre synthesis. Design your script and .sdc file such that it generates optimized reports of timing.

2. Perform the complete physical design flow:

- Floor planning
 - Choose core utilization, aspect ratio, offsets appropriately. Make proper IO file.
 - Justify every chosen dimension and parameter.
- Power Planning
 - Add power rings and stripes (VDD/VSS).
 - Ensure proper PDN network.
- Placement
 - Perform standard cell placement.
 - Eliminate overlaps and check placement legality.
- Clock Tree Synthesis (CTS)
 - Perform pre and post CTS.
 - Avoid clock gating issues.
- Routing
 - Route all signals and power nets.
 - Ensure minimal routing congestion.
- Post-Route Optimization
 - Ensure no DRC, LVS, or timing violations.
- Timing Analysis
 - Perform setup and hold analysis.
 - Achieve closure at target frequency with no WNS/TNS violations.
- GDSII Generation
 - Generate the final GDSII layout file.
 - Check connectivity and DRC checks.
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- Report Requirements
 - The report must include:

- Snapshots from every step (floorplan, placement, CTS, routing, etc.)
 - Detailed explanation of all chosen dimensions and parameters
 - Comments on each tool-generated and user-written file
 - All timing/area/power reports with interpretation
 - Final metrics summary (area, timing, frequency, utilization, congestion map)
 - Lessons learned, challenges faced, and how they were resolved.
- Design Constraints
 - No DRC or timing violations are allowed.
 - Final design must run at target frequency without setup/hold violations.
 - Minimize area without causing routing congestion or violating timing.
 - You can include different constraints in your constraint file such as max fanout, clock gating, power gating for better results.